

# Mathematics as Stabilized Information

A Discussion Diary on Human Practice, Precision, Education, and AI

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**The guiding question**

How do physical systems - human brains, communities, written media, and artificial intelligence - acquire, stabilize, communicate, and use mathematical concepts?

**The central wager**

Mathematics should first be studied as a real activity inside civilization. Its unusual exactness does not separate it from information technology; rather, it may represent one of the most demanding forms of information stabilization that human societies have built.

Discussion recorded on August 29, 2026

This diary records a developing research program, not a set of settled conclusions.

## Opening: Why begin with human activity?

**Beginning of the discussion.** I did not want to begin with the usual ontological question: *Do numbers, sets, or geometric objects exist independently of us?* That question may eventually matter, but it seemed too detached from the phenomenon I actually wanted to understand. Mathematics is not encountered first as an abstract realm. It appears in history as something that people do.

Mathematicians live inside societies. They use languages, diagrams, instruments, schools, books, institutions, and technologies. Their work is entangled with philosophy, astronomy, construction, navigation, trade, physics, and engineering. In ancient Greece, the figures whom we now separate into philosophers, mathematicians, scientists, and engineers often belonged to a single intellectual culture. The later division into disciplines should not be projected backward as though it were fundamental.

This suggested a different starting point for a philosophy of mathematics:

### Working thesis

Before asking what mathematical objects are in themselves, study how mathematical practices arise, become reliable, spread through a community, and acquire the power to describe or control parts of the world.

This approach is “more scientific” not because philosophy is rejected, but because the primary evidence is actual activity: counting, measuring, drawing, proving, teaching, calculating, designing, predicting, and controlling. Mathematics is treated as a phenomenon in human civilization.

## Information technology in the broadest sense

The phrase *information technology* usually evokes computers, digital storage, or Shannon’s theory. That was not the intended meaning here. The term was being used in a much broader and more primitive sense.

A physical system functions as an information technology when we can prepare it in distinguishable states, preserve those states long enough, and allow another system to detect or decode them. Under this description, a sheet of paper is an information technology. So is a blackboard. So is spoken language. A controlled sequence of sound waves can carry a pattern from one nervous system to another. Writing converts fleeting sound or thought into a more durable physical configuration.

The example of the word *apple* made this vivid. Different people write the letters with different shapes. A handwritten *p* need not be geometrically identical across inscriptions. Yet a community recognizes the relevant pattern, the order of the letters, and the word. The code is stable not because every physical mark is identical, but because some variations are ignored while certain relations are preserved.

Natural language is therefore an information technology in this broad sense. Mathematics can also be spoken, but written notation, diagrams, and symbolic manipulation carry structures that are difficult to preserve in speech alone. The contrast is not between “language” and “technology.” Language itself is a family of embodied technologies for producing reproducible states in other minds.

**Provisional operational definition**

An information technology is any organized physical process by which a state can be prepared, made sufficiently stable or repeatable, detected by another system, and used to guide further activity.

Under this definition, biological perception also belongs to the story. Eyes, ears, skin, and nervous systems transform physical differences - light, pressure, temperature, vibration - into patterns that affect behavior. The brain may be viewed as an information-processing technology produced by biological evolution rather than deliberate human engineering.

This does not yet tell us what information ultimately *is*. It provides a practical orientation: look at the preparation, transmission, decoding, use, and stabilization of physical differences.

**Paper, ruler, compass, and the production of precision**

The discussion then returned to ancient geometry. Euclidean geometry did not emerge in a vacuum. It depended on practical means of drawing and preserving forms: surfaces, straightedges, compasses, marks, and shared standards of acceptable construction.

A line drawn with a ruler is never a perfectly one-dimensional mathematical line. It has width, texture, irregularities, and a finite physical resolution. If magnified far enough, the paper, graphite, and ruler cease to resemble the smooth idealization. Atomic structure and quantum physics eventually become relevant. A pencil mark is not literally a geometric point, and a drawn segment is not literally an ideal straight line.

Yet this does not make geometry meaningless. It reveals that physical exactness and mathematical exactness are different. A geometrical practice can tolerate some physical deviation while still enforcing a stricter invariant: the mark must be straight enough for the intended operations and checks. A community can put a ruler against the mark, compare constructions, and decide whether the relevant relation has been preserved.

**Working thesis**

Mathematical precision is not the physical perfection of a mark. It is the reproducibility of selected distinctions and transformations despite irrelevant physical variation.

The scale matters. A construction may count as exact within the resolution of its instruments and purposes. But mathematics also performs a remarkable move: it separates the abstract relation from every particular instrument. Once the concept of a line or point has been stabilized, it can be reused in contexts that ancient drawing technologies never anticipated.

**The physical point and the abstract point**

A pencil dot occupies an area. An abstract point does not. Nevertheless, the abstract notion continues to be useful in very different theories. In solid-state physics, atoms may be idealized as points on a lattice. In quantum mechanics, one speaks of amplitudes associated with positions or localized states, even though the physical interpretation is far removed from a dot on paper.

The same abstract vocabulary can migrate across scales and theories. This portability is astonishing. The original representational technology may have helped stabilize the concept, but the mature abstraction is no longer tied to that technology.

One way to describe idealization is as a disciplined loss of information. Thickness, roughness, atomic structure, and manufacturing defects are suppressed so that incidence, order, distance, or continuity can be treated with greater logical clarity. The abstraction is useful because it ignores many physical details while preserving the relations needed for reasoning.

#### Question left open

Why are some abstractions so portable? Why can a concept first stabilized through macroscopic drawing practices later organize theories of microscopic matter, computation, or high-dimensional structure?

## From static marks to dynamic concepts

A symbol on paper may be physically stable for hours or centuries. A mathematical concept in a brain is not stable in the same way. Neural activity is dynamic, distributed, context-sensitive, and constantly changing. The concept of a point is not a tiny static object stored at a fixed location in the head.

This forced a refinement. Stability should not always mean persistence of an identical physical state. In cognition, stability may instead mean the reliable ability to *re-enter* a pattern of activity.

When a trained mathematician encounters a familiar concept, the relevant inferential habits become available. The mathematician can recognize examples, perform standard operations, notice violations, connect the concept with related results, and reconstruct arguments. The exact neural state need not repeat. What repeats is a structured capacity.

#### Provisional operational definition

For an individual intelligent system, a concept is stabilized when suitable cues reliably reactivate a sufficiently similar family of operations, expectations, distinctions, and inferences.

This is a practical definition. It avoids asking whether a concept is a single inner picture. A concept may be better understood as a reproducible ability: a way of navigating a space of possible reasoning.

The word *line* illustrates this. An analyst may immediately activate limits, completeness, Cauchy sequences, order, and  $\varepsilon$ - $\delta$  arguments. A point-set topologist may foreground neighborhoods, connectedness, separation, or compactification. A homotopy theorist may use the interval as a parameter object for deformations. These are not wholly unrelated concepts, but neither are they identical mental images. They are overlapping inferential repertoires shaped by different practices.

The important point is not that every mathematician secretly means something private and incommensurable. With training, a person can move between these repertoires. Concepts are learnable patterns of operation, not immutable pictures permanently attached to individuals.

## A hierarchy of stability

By the middle of the discussion, the word *stability* had acquired several distinct meanings. Keeping them separate became essential.

| Level                               | What is stabilized?                                                                                                                                               |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Physical persistence                | A mark, sound pattern, memory trace, file, or device state remains detectable or reconstructible for a relevant period.                                           |
| Intrapersonal reactivation          | The same person can later re-enter a similar pattern of reasoning and use the concept consistently.                                                               |
| Pedagogical reproducibility         | An ordinary person with suitable background can acquire the concept through a reasonable period of training.                                                      |
| Communal coordination               | Different members of a community can communicate, check, criticize, and extend reasoning involving the concept.                                                   |
| Cross-architecture interoperability | Different kinds of intelligent systems, such as humans and AIs, can exchange mathematically useful inputs and outputs even when their internal primitives differ. |

These levels should not be collapsed. A private notation can be physically stable and meaningful to one person without becoming a communal mathematical concept. Conversely, a communal concept can be instantiated by different neural processes in different people while remaining operationally shared.

A concept becomes public not because every brain contains an identical state, but because training and communication align the actions that matter. Participants can make compatible inferences, detect similar errors, and use the concept to cooperate.

#### Working thesis

Public mathematical stability is not identity of internal representation. It is reliable coordination of mathematical performance across persons, times, and media.

## Education as a technology of conceptual stabilization

The difference between a textbook and a research paper supplied an important example. A textbook written for beginners tries to make each major inferential step reconstructible by a reader who lacks the field's mature intuition. A research paper assumes a different stabilized state. It omits transitions that an expert is expected to supply, activates familiar patterns with compressed notation, and relies on a shared sense of what is routine.

This led to an operational criterion for an abstract mathematical concept. The criterion should not be calibrated to rare geniuses, exceptional visualizers, or people who report obtaining formulas in dreams. It should concern ordinary intelligent learners under reasonable professional training.

#### Provisional operational definition

A mathematical concept is humanly stabilizable when an ordinary learner with appropriate prerequisites can, within a reasonable period of training, reliably recognize the concept, perform its basic operations, reconstruct its standard inferences, communicate about it, and transfer it to new examples.

This definition is deliberately graded. "Reasonable time" depends on the complexity of the field,

the learner's prerequisites, and available educational technologies. But the criterion makes the idea testable in principle. We can ask how much instruction is required, how robust the acquired capability is, how far it transfers, and which media make the learning process more efficient.

Education is therefore not merely the transmission of already completed mathematics. It is an information technology for producing reproducible cognitive capacities. Definitions, examples, exercises, diagrams, notation, proof styles, and mentorship all participate in stabilizing a concept inside new minds.

The most revealing sign of stabilization may be what no longer needs to be said. Once a pattern becomes fluent, intermediate steps can be recovered rather than explicitly supplied. Expert compression is possible because a community has invested in shared training.

## Mathematics as the high-precision case

The broader project concerns science and understanding, not mathematics alone. Animals, bacteria, nervous systems, scientific communities, and artificial agents all respond to their environments and may construct internal states that support prediction or control. Mathematics is special within this larger picture because its norms of stability are unusually demanding.

This does not mean that every mathematical statement corresponds to a physically exact measurement. Nor does it mean that other sciences are imprecise. The claim is subtler: mathematical communities organize symbols, definitions, and transformations so that disagreements can often be localized and inferential steps can, in principle, be reproduced with exceptional exactness.

Mathematical accuracy includes several dimensions:

- **Syntactic precision:** symbols and expressions can be distinguished and manipulated according to explicit rules.
- **Inferential reproducibility:** a proof can be checked by another trained person or, increasingly, by a formal system.
- **Conceptual discipline:** definitions constrain which operations and analogies are legitimate.
- **Composability:** stabilized results can be used as components in longer arguments without reopening every earlier step.
- **Transportability:** abstract structures can be realized in different physical, scientific, and mathematical settings.

Thus mathematics may be understood as a highly refined information practice: one that makes reasoning unusually portable, compressible, and checkable.

### Question left open

Can this special precision be characterized quantitatively? For example, by robustness under re-encoding, error-correction, proof reconstruction, transfer to new problems, or the amount of training needed to reproduce an inference?

## AI training, fast inference, and nonhuman concepts

The discussion then shifted from human education to artificial intelligence. There is an obvious structural analogy: training can be long and expensive, while a trained system may produce an answer in seconds. Humans also undergo years of education and later perform familiar inferences rapidly. The analogy is incomplete, but it exposes a common problem: how does prolonged interaction with data and feedback reorganize a system so that a family of responses becomes immediately available?

The deeper possibility is that an AI system may stabilize mathematical concepts that are not humanly stabilizable. After training on enormous bodies of mathematics, a model might develop internal structures that support reliable reasoning, compression, or discovery, while no human can acquire the same primitive structures through any reasonable curriculum.

This possibility requires a careful distinction:

- The AI may possess a stable internal capability.
- Humans may observe consistent inputs and outputs.
- Humans may verify proofs, predictions, or consequences.
- Yet humans may fail to develop the AI's own primitive intuition.

Would that count as mathematical understanding? The answer depends on the operational criteria we adopt. If understanding requires shared human intuition, then no. If it requires robust use, generalization, error correction, and explanatory compression relative to the system's own architecture, then perhaps yes.

The diary therefore arrived at a relative notion:

#### Working thesis

Conceptual stability is relative to a class of intelligent systems and to a training process. A concept may be stable for humans, stable for AIs, stable for both, or stable for neither.

This does not imply that every opaque model state deserves the name "concept." Mere memorization or accidental correlation is insufficient. The stronger evidence would include systematic generalization, compositional use, counterfactual sensitivity, transfer between problems, and the ability to repair reasoning after errors.

#### Human intuition after AI discovery

Even when humans cannot internalize an AI's primitive concept, they may develop a second-order intuition around its behavior. Scientists already work with systems that cannot be directly visualized. We use projections, analogies, invariants, effective theories, and experimental signatures.

A future mathematical community might interact with an AI-discovered structure through examples, queries, theorems, visualizations, compressed explanations, and verified consequences. Over time, humans could stabilize a different concept that tracks the same underlying regularities without reproducing the AI's internal representation.

The human and AI concepts would then be related through a translation interface rather than identity. Their intuitions could differ while their mathematical actions remain coordinated.

## The interface becomes part of mathematics

The phrase *user interface* became unexpectedly important. If two intelligences cannot share primitives, the practical problem is to build an interface through which they can still do mathematics together.

A good mathematical interface would not merely display answers. It would expose enough structure for humans to test, challenge, navigate, and reuse the system's results. It might provide:

- human-readable conjectures and definitions;
- formally checkable proofs;
- examples and counterexamples selected for pedagogy;
- multiple representations of the same structure;
- uncertainty estimates and failure conditions;
- interactive explanations adapted to the user's current conceptual repertoire;
- tools for translating between human and machine-native abstractions.

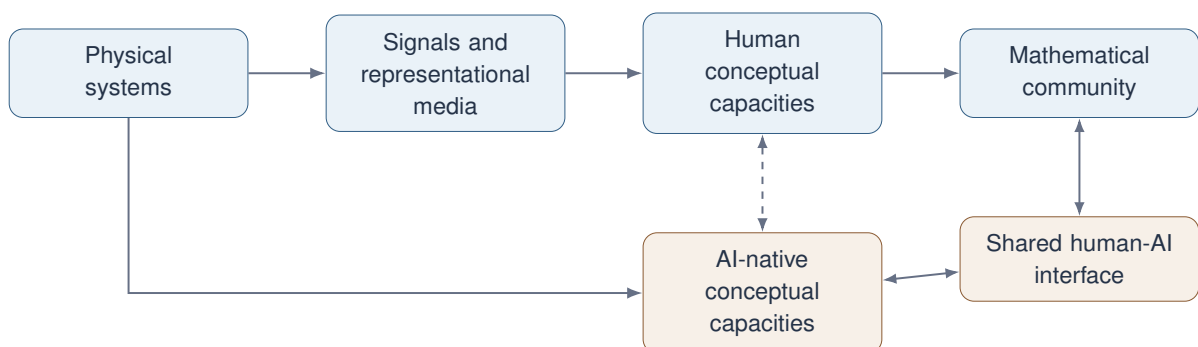
The interface is not a superficial layer added after discovery. It may become the place where a new communal mathematics is formed. A machine-native concept becomes part of human science only when there are stable practices for interrogating, validating, and composing its consequences.

### Provisional operational definition

Cross-system mathematical understanding is achieved when distinct kinds of intelligent systems can coordinate reliable mathematical action through an interface, even if they do not share the same internal representations.

This returns to the beginning of the conversation. Paper, diagrams, speech, notation, textbooks, formal proof assistants, and AI interfaces are all media that coordinate intelligent systems. The history of mathematics can be read as the history of increasingly powerful interfaces for stabilizing and sharing patterns of reasoning.

## A conceptual map of the discussion



The diagram should not be read as a linear pipeline. Every arrow also points backward through feedback. Instruments reshape perception. Writing reshapes memory. Communities redesign

notation. AI systems alter education. New mathematical concepts enable new technologies, which in turn alter the concepts that can be stabilized.

## What we arrived at today

By the end of the conversation, the original topic - philosophy of mathematics - had expanded into a general inquiry about understanding in physical systems. Mathematics remained central because it is a particularly strict test case.

The following theses emerged:

1. Mathematics should initially be studied as an activity inside civilization, not as a completed realm detached from history, bodies, media, or institutions.
2. Information technology, in the broad sense, includes any physical means of preparing, preserving, detecting, and using distinguishable states.
3. Spoken language, writing, diagrams, rulers, compasses, nervous systems, schools, and AI models all belong to this extended technological ecology.
4. Mathematical exactness is not literal physical perfection. It is the disciplined preservation of selected distinctions and transformations across variable physical realizations.
5. An abstract concept in a mind is not a static neural object. It is a reproducible capacity to reactivate a structured pattern of reasoning.
6. A communal mathematical concept is one that ordinary qualified learners can acquire in reasonable time and use in mutually checkable ways.
7. Education is a technology for stabilizing such capacities across generations.
8. Different kinds of intelligent systems may stabilize different primitive concepts.
9. Human-AI collaboration may depend less on shared internal intuition than on reliable, inspectable, and improvable interfaces.
10. Future mathematics may include machine-native structures for which humans develop only indirect or second-order intuitions.

## Questions for the next entry

Several questions now deserve more precise treatment:

### Research questions

1. What operational tests distinguish a genuine concept from memorized input-output behavior?
2. How should “reasonable training time” be defined for humans and for artificial systems?
3. Can conceptual stability be measured through transfer, robustness, compression, compositionality, or error correction?
4. What kinds of external media made particular mathematical abstractions historically possible?
5. When two systems use different internal primitives, what makes a translation interface faithful?
6. Could an AI-generated concept be mathematically fruitful but permanently resistant to human stabilization?
7. If humans build a second-order intuition around such a concept, in what sense are both systems understanding the same thing?
8. How would the institutions of proof, education, authorship, and explanation change in a mathematics no longer limited by human cognitive architecture?

## Closing reflection

The conversation began with paper, pencils, rulers, and ancient geometry. It ended with the possibility of mathematical ideas native to artificial minds. The path between them was not accidental. Both ends concern the same problem: how a physical system can carry a pattern that another system can recover and use.

A line on paper, a definition in a textbook, a habit in a mathematician’s mind, a proof shared by a community, and a representation inside an AI are not the same kind of object. But each can participate in a chain of stabilization. Mathematics may be the civilizational practice that pushes this chain toward its highest currently known demands of precision, reproducibility, and composability.

The long-term project is therefore larger than asking whether AI can solve human mathematics. It asks how mathematics changes when the systems capable of stabilizing concepts are no longer all human, and how different intelligences might build a common scientific world without sharing the same inner forms of thought.